

ODESA, Ukraine

WMO: 338370

Lat: 46.4408N

Lon: 30.7703E

Elev: 42

StdP: 100.82

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-10.3	-16.8	0.9	-12.7	-13.9	1.1	-9.7	13.2	-4.3	11.6	-3.1	4.6	20	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	32.9	20.6	31.0	20.2	29.3	19.9	23.1	27.9	22.2	27.3	21.4	26.6	3.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	16.5	25.4	20.8	15.5	24.7	19.8	14.6	24.0	68.8	27.9	65.5	27.2	62.6	26.4	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	8.9	8.0	DB	-17.0	35.7	2.6	2.2	-18.9	37.2	-20.5	38.5	-21.9	39.7	-23.9	41.2	(4)
			WB	-17.4	25.1	2.6	1.1	-19.3	25.9	-20.8	26.5	-22.2	27.2	-24.1	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-1.1	0.3	4.3	10.4	16.4	20.9	23.6	23.2	17.5	11.3	6.2	1.0	(6)
(7)		DBStd	9.55	4.98	4.92	3.98	3.47	3.45	3.12	2.78	3.21	3.52	4.09	4.55	4.96	(7)
(8)		HDD10.0	1271	344	273	178	35	1	0	0	1	33	128	278		(8)
(9)		HDD18.3	3050	602	506	434	240	80	11	1	2	56	220	363	536	(9)
(10)		CDD10.0	1720	0	0	3	46	199	328	421	409	226	73	16	1	(10)
(11)		CDD18.3	457	0	0	0	1	20	88	163	153	31	2	0	0	(11)
(12)		CDH23.3	3737	0	0	0	9	137	667	1424	1329	167	3	0	0	(12)
(13)		CDH26.7	1212	0	0	0	2	24	183	488	485	30	0	0	0	(13)
(14)	Wind	WSAvg	3.8	4.1	4.2	4.3	3.8	3.4	3.4	3.3	3.5	3.7	4.0	4.0	(14)	
(15)	Precipitation	PrecAvg	460	41	32	32	28	37	51	45	41	38	38	37	(15)	
(16)		PrecMax	720	113	90	78	65	100	107	122	138	116	172	106	120	(16)
(17)		PrecMin	341	7	1	0	0	2	13	2	2	0	4	0	1	(17)
(18)		PrecStd	100	29	22	22	18	26	23	32	36	36	34	25	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	12.2	17.9	24.3	29.2	32.2	35.0	35.9	29.8	23.2	18.0	12.9	(19)
(20)		0.4%	MCWB	7.1	8.4	11.0	15.7	18.2	20.1	20.5	21.1	18.4	16.7	13.8	10.6	(20)
(21)		2%	DB	8.1	9.2	14.1	20.2	26.8	30.4	32.9	33.1	27.0	21.1	15.6	11.0	(21)
(22)		2%	MCWB	6.6	6.7	8.7	13.0	18.0	20.0	20.9	20.5	17.7	16.7	13.7	9.5	(22)
(23)		5%	DB	6.7	7.7	11.8	18.1	24.8	28.9	31.0	31.2	25.1	19.1	14.0	9.4	(23)
(24)		5%	MCWB	5.7	5.9	7.6	11.8	17.2	19.6	20.5	20.1	17.7	15.7	12.4	8.5	(24)
(25)		10%	DB	5.1	6.1	9.9	16.0	22.8	27.1	29.2	29.2	23.3	17.5	12.8	8.0	(25)
(26)		10%	MCWB	4.3	5.0	6.6	10.9	16.3	19.1	20.2	19.8	17.4	14.6	11.5	7.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	9.1	11.3	16.2	20.6	23.4	24.2	24.7	22.1	19.2	15.4	11.4	(27)
(28)		0.4%	MCDWB	9.1	11.9	16.5	22.5	25.0	28.4	28.1	29.4	24.8	21.2	16.8	12.4	(28)
(29)		2%	WB	6.7	7.2	9.5	14.1	19.3	22.1	23.2	23.1	20.7	17.7	14.0	10.0	(29)
(30)		2%	MCDWB	7.5	8.5	13.5	18.6	24.0	26.8	27.9	28.0	23.5	19.8	15.4	10.5	(30)
(31)		5%	WB	5.6	6.1	8.1	12.8	18.3	21.2	22.3	22.1	19.5	16.5	12.8	8.5	(31)
(32)		5%	MCDWB	6.3	7.1	10.6	16.5	23.0	25.7	27.5	27.4	22.9	18.6	13.9	9.3	(32)
(33)		10%	WB	4.3	5.0	7.0	11.6	17.3	20.3	21.5	21.2	18.4	15.0	11.6	7.2	(33)
(34)		10%	MCDWB	5.0	5.9	8.8	15.2	21.6	25.0	27.0	26.6	21.9	17.0	12.7	7.9	(34)
(35)	Mean Daily Temperature Range		MDBR	5.7	6.4	7.7	8.8	9.8	10.0	10.4	10.3	9.5	8.1	6.3	5.9	(35)
(36)		5% DB	MCDBR	6.9	8.5	11.1	11.7	12.4	11.9	12.0	12.2	11.5	10.0	7.5	7.0	(36)
(37)			MCWBR	5.6	6.2	6.9	6.3	5.9	5.3	4.8	5.0	6.0	6.3	5.5	5.7	(37)
(38)			MCDBR	6.4	7.8	9.6	10.5	11.1	10.3	10.4	10.5	9.7	8.5	6.3	6.3	(38)
(39)		5% WB	MCWBR	5.5	6.1	6.5	6.1	5.8	5.4	5.1	5.4	6.2	6.1	5.0	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.343	0.370	0.398	0.443	0.419	0.417	0.444	0.444	0.413	0.382	0.361	0.341	(40)	
(41)		taud	2.314	2.260	2.227	2.127	2.288	2.326	2.261	2.248	2.298	2.358	2.347	2.340	(41)	
(42)		Ebn at Noon	729	784	818	812	850	852	821	804	795	764	703	687	(42)	
(43)		Edh at Noon	79	101	120	145	128	124	131	127	111	91	76	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.02	1.93	3.27	4.96	6.34	6.69	6.67	5.94	4.19	2.59	1.23	0.87	(44)	
(45)		RadStd	0.18	0.26	0.31	0.41	0.48	0.40	0.36	0.31	0.46	0.29	0.19	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.69	N/A	N/A	+1.03	+0.47	+0.42	-115	-153	+119	+83	(46)
(47)	Regional (9 neighbors)	+0.63	N/A	N/A	+0.79	N/A	N/A	-101	-141	+109	+76	(47)

Nomenclature: See separate page